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Qualification Pack



Junior Engineer -Traction (Metro Rail)

QP Code: DMR/LSC/Q0101

Version: 1.0

NSQF Level: 4.5

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DMR/LSC/Q0101: Junior Engineer -Traction (Metro Rail)

Brief Job Description

The Junior Engineer - Traction oversees the installation, operation, and maintenance of Power Supply Installation (PSI) and SCADA systems, ensuring reliable power supply within the traction network. This role includes coordinating Overhead Electrification (OHE) maintenance, coordinating with technical teams, and ensuring adherence to safety protocols for efficient traction system operations.

Personal Attributes

The individual should have strong analytical skills, attention to detail, and effective communication to oversee complex traction systems. A proactive approach, technical aptitude, and the ability to lead teams under high-pressure situations are essential for smooth operations and maintenance.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [DMR/LSC/N9901: Implement metro operations training and safety procedures](#)
2. [DMR/LSC/N0101: Oversee Power Supply Installation \(PSI\) and maintenance in the traction system](#)
3. [DMR/LSC/N0102: Oversee power system protection and control](#)
4. [DMR/LSC/N0103: Oversee the SCADA system operations and maintenance](#)
5. [DMR/LSC/N0104: Oversee the maintenance of Overhead Electrification \(OHE\) systems](#)
6. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Logistics
Sub-Sector	Transport (Metro Rail)
Occupation	Traction
Country	India
NSQF Level	4.5
Credits	17

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/3122.5600
Minimum Educational Qualification & Experience	Diploma (3 years Engineering Diploma / equivalent in Electrical / Mechanical or in relevant field)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-4.5-TW-03511-2025-V1-DMRCA
NQR Version	1.0

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DMR/LSC/N9901: Implement metro operations training and safety procedures

Description

This unit focuses on implementing structured training and safety measures for metro operations, ensuring technical knowledge, system readiness, and emergency preparedness.

Scope

The scope covers the following :

- Training Module Overview
- Safety Overview
- Store Management and Procurement
- Overview of Telecom, Signaling, and Track Systems
- Overview of Traction, SCADA, and E&M Systems
- OCC and Rolling Stock Overview
- Operations and Customer Care
- Practical Visits
- First Aid Training
- Fire Fighting Systems
- Security in Metro Systems
- Metro Rail General Rules 2020
- Safety Circulars and Contingency Plans

Elements and Performance Criteria

Training Module Overview

To be competent, the user/individual on the job must be able to:

- PC1.** follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.
- PC2.** conduct identity verification and maintain attendance records
- PC3.** ensure adherence to training timelines and program deliverables

Safety Overview

To be competent, the user/individual on the job must be able to:

- PC4.** identify the role of the safety department and follow measures for personal, passenger, and system safety
- PC5.** use appropriate Personal Protective Equipment (PPE) in all designated areas
- PC6.** follow the hazard identification and risk assessment processes

Store Management and Procurement

To be competent, the user/individual on the job must be able to:

- PC7.** perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.
- PC8.** monitor inventory levels and ensure timely requisitions for required materials
- PC9.** maintain records of procurement processes and vendor communication

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Overview of Telecom, Signaling, and Track Systems

To be competent, the user/individual on the job must be able to:

- PC10.** identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks
- PC11.** identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.
- PC12.** identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks
- PC13.** report malfunctions in telecom, signaling, or track systems using the correct reporting protocols

Overview of Traction, SCADA, and E&M Systems

To be competent, the user/individual on the job must be able to:

- PC14.** identify traction power and SCADA components and interpret sectioning diagrams
- PC15.** identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality
- PC16.** report unusual system performance and assist in routine inspections

OCC and Rolling Stock Overview

To be competent, the user/individual on the job must be able to:

- PC17.** familiarize with OCC systems, tools, and the role of OCC controllers
- PC18.** identify rolling stock components and their maintenance procedures
- PC19.** report rolling stock irregularities observed during operations

Operations and Customer Care

To be competent, the user/individual on the job must be able to:

- PC20.** familiarize with station operations, revenue management, and crew control functions
- PC21.** identify public complaint channels, possible causes, and resolve complaints using proper processes
- PC22.** assist in managing passenger inquiries and ensuring customer satisfaction

Practical Visits

To be competent, the user/individual on the job must be able to:

- PC23.** conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations
- PC24.** document key observations during practical visits for knowledge reinforcement
- PC25.** identify operational challenges during visits and suggest improvement areas

First Aid Training

To be competent, the user/individual on the job must be able to:

- PC26.** identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)
- PC27.** follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness
- PC28.** perform artificial respiration, CPR, and safe transportation of injured persons using stretchers
- PC29.** participate in simulated first aid scenarios to enhance preparedness

Fire Fighting Systems

To be competent, the user/individual on the job must be able to:

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- PC30.** identify elements of combustion, classes of fire, and appropriate extinguishing methods
- PC31.** operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants
- PC32.** conduct practical drills and identify fire suppression equipment at metro stations
- PC33.** respond to fire incidents promptly, ensuring passenger and system safety

Security in Metro Systems

To be competent, the user/individual on the job must be able to:

- PC34.** identify security threats, red alert situations, and follow protocols for disaster management
- PC35.** handle theft cases, manage abandoned persons, and lodge FIRs as per protocol
- PC36.** coordinate with security personnel to mitigate security breaches effectively

Metro Rail General Rules 2020

To be competent, the user/individual on the job must be able to:

- PC37.** interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols
- PC38.** follow operational procedures for maintenance, station control duties, and driverless train systems
- PC39.** ensure compliance with rules for safe and efficient train operations

Safety Circulars and Contingency Plans

To be competent, the user/individual on the job must be able to:

- PC40.** implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation
- PC41.** follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises
- PC42.** ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods
- PC43.** manage degraded mode operations, including train movement during interlocking failures
- PC44.** oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections
- PC45.** facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks
- PC46.** execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras
- PC47.** oversee train stabling, depot entry/exit, train movement, and fitness checks
- PC48.** conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detainment of passengers due to technical failure, explosion, or fire
- PC49.** operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train
- PC50.** coordinate vehicle movement on link lines and rescue immobilized trains
- PC51.** respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety
- PC52.** operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of punctuality, discipline, and structured training
- KU2.** the procedures for trainee verification and attendance record maintenance
- KU3.** the role of time management in achieving training milestones
- KU4.** the role of the safety department and station/depot safety protocols
- KU5.** the use and importance of PPE
- KU6.** the basics of hazard identification, risk management, and accident reporting
- KU7.** the basics of stores management (inventory control, documentation)
- KU8.** the procurement processes and coordination across departments
- KU9.** the procedures for vendor management and maintaining procurement transparency
- KU10.** the telecom components, including AFC, CCTV, clocks, and communication systems
- KU11.** the basics of signaling systems (CATC, CBTC) and reporting unusual signal observations
- KU12.** the key P-Way components, track patrolling, and reporting track conditions
- KU13.** the common malfunctions in telecom, signaling, and track systems
- KU14.** the traction power systems and sectioning diagrams
- KU15.** the importance of SCADA and its functionalities
- KU16.** the role of E&M systems (DG sets, UPS) and their use during emergencies
- KU17.** the key preventive maintenance checks for E&M systems
- KU18.** the OCC systems, tools, and controller responsibilities
- KU19.** the rolling stock components, features, and basic maintenance
- KU20.** the reporting and addressing of common rolling stock issues
- KU21.** the station operations, revenue management, and crew coordination
- KU22.** the causes of public complaints and grievance resolution processes
- KU23.** the techniques for effective customer service and passenger engagement
- KU24.** the observations and learning outcomes from OCC, station, depot, and tunnel visits
- KU25.** the identifying real-world challenges during visits and linking theory to practice
- KU26.** the basic principles of first aid and key human body functions
- KU27.** the management of bleeding, fractures, burns, and unconsciousness
- KU28.** the artificial respiration, CPR, and safe patient transportation
- KU29.** the importance of first aid preparedness during emergencies
- KU30.** the elements of combustion, fire classifications, and extinguishing methods
- KU31.** the principles of passive/active fire protection systems
- KU32.** the practical usage of fire extinguishers, hydrants, and drills
- KU33.** the emergency response procedures during fire incidents
- KU34.** the security protocols, red alert situations, and theft management processes
- KU35.** the protocols for managing abandoned persons and disaster situations
- KU36.** the role of security personnel in ensuring passenger and asset safety
- KU37.** the key terminologies, signal usage, and operational protocols

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- KU38.** the rules for maintenance works, train operations, and driverless train systems
- KU39.** the role of regulatory compliance in metro rail safety
- KU40.** the safety circulars and contingency plans for fire/security incidents
- KU41.** the importance of proactive contingency planning for emergency scenarios.
- KU42.** the importance role-playing and drills to improve safety and response performance
- KU43.** the procedures for TSR imposition, axle counter reset, Train ID management, and CATS operations
- KU44.** the engineering possession protocols for maintenance tasks during non-revenue hours, including solar system testing and hot work
- KU45.** the guidelines for safely halting train operations during emergencies, natural disasters, or technical failures
- KU46.** the steps for managing degraded mode operations and interlocking failures
- KU47.** the SOP and SWO/DWO lifecycle management, including revisions and timetable trials for new sections
- KU48.** the ATO, UTO, and ATB mode operations, emphasizing operational readiness and AFTC checks
- KU49.** the procedures for neutral section negotiation, pantograph damage management, and monitoring Panto-OHE interfaces using cameras
- KU50.** the best practices for train stabling, depot operations, and train fitness verification
- KU51.** the emergency evacuation protocols, focusing on passenger safety in UG and mid-tunnel situations
- KU52.** the guidelines for operating non-ATP vehicles, handling rail fractures, weld failures, and resetting alarms during rollbacks
- KU53.** the vehicle operations on link lines and immobilized train rescue procedures
- KU54.** the protocols for fire/smoke incidents, particularly UG fires, and ensuring safety during such incidents
- KU55.** the operation of ETS/BLS plungers and removal of foreign objects from OHE systems, along with OHE safety practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** convey clear and concise information to teams, junior engineers, and customers
- GS2.** effectively prioritize tasks and adhere to training schedules and operational timelines
- GS3.** identify operational challenges and propose practical solutions during training or real-time operations
- GS4.** assess system performance, identify malfunctions, and report anomalies accurately
- GS5.** work cohesively with peers, instructors, and junior engineers to meet training objectives
- GS6.** address passenger queries and manage grievances professionally
- GS7.** proactively identify risks, follow safety protocols, and contribute to hazard-free operations
- GS8.** adapt to diverse training environments, system layouts, and evolving operational scenarios
- GS9.** accurately maintain logs, records, and reports for training and practical activities

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- GS10.** monitor systems, equipment, and processes, identifying areas requiring intervention or improvement

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Training Module Overview</i>	2	2	-	1
PC1. follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.	-	-	-	-
PC2. conduct identity verification and maintain attendance records	-	-	-	-
PC3. ensure adherence to training timelines and program deliverables	-	-	-	-
<i>Safety Overview</i>	2	2	-	1
PC4. identify the role of the safety department and follow measures for personal, passenger, and system safety	-	-	-	-
PC5. use appropriate Personal Protective Equipment (PPE) in all designated areas	-	-	-	-
PC6. follow the hazard identification and risk assessment processes	-	-	-	-
<i>Store Management and Procurement</i>	2	2	-	1
PC7. perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.	-	-	-	-
PC8. monitor inventory levels and ensure timely requisitions for required materials	-	-	-	-
PC9. maintain records of procurement processes and vendor communication	-	-	-	-
<i>Overview of Telecom, Signaling, and Track Systems</i>	2	2	-	1
PC10. identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks	-	-	-	-
PC11. identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks	-	-	-	-
PC13. report malfunctions in telecom, signaling, or track systems using the correct reporting protocols	-	-	-	-
<i>Overview of Traction, SCADA, and E&M Systems</i>	2	2	-	1
PC14. identify traction power and SCADA components and interpret sectioning diagrams	-	-	-	-
PC15. identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality	-	-	-	-
PC16. report unusual system performance and assist in routine inspections	-	-	-	-
<i>OCC and Rolling Stock Overview</i>	2	2	-	1
PC17. familiarize with OCC systems, tools, and the role of OCC controllers	-	-	-	-
PC18. identify rolling stock components and their maintenance procedures	-	-	-	-
PC19. report rolling stock irregularities observed during operations	-	-	-	-
<i>Operations and Customer Care</i>	2	2	-	1
PC20. familiarize with station operations, revenue management, and crew control functions	-	-	-	-
PC21. identify public complaint channels, possible causes, and resolve complaints using proper processes	-	-	-	-
PC22. assist in managing passenger inquiries and ensuring customer satisfaction	-	-	-	-
<i>Practical Visits</i>	1	1	-	0.5
PC23. conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. document key observations during practical visits for knowledge reinforcement	-	-	-	-
PC25. identify operational challenges during visits and suggest improvement areas	-	-	-	-
<i>First Aid Training</i>	1	1	-	0.5
PC26. identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)	-	-	-	-
PC27. follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness	-	-	-	-
PC28. perform artificial respiration, CPR, and safe transportation of injured persons using stretchers	-	-	-	-
PC29. participate in simulated first aid scenarios to enhance preparedness	-	-	-	-
<i>Fire Fighting Systems</i>	1	1	-	0.5
PC30. identify elements of combustion, classes of fire, and appropriate extinguishing methods	-	-	-	-
PC31. operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants	-	-	-	-
PC32. conduct practical drills and identify fire suppression equipment at metro stations	-	-	-	-
PC33. respond to fire incidents promptly, ensuring passenger and system safety	-	-	-	-
<i>Security in Metro Systems</i>	1	1	-	0.5
PC34. identify security threats, red alert situations, and follow protocols for disaster management	-	-	-	-
PC35. handle theft cases, manage abandoned persons, and lodge FIRs as per protocol	-	-	-	-
PC36. coordinate with security personnel to mitigate security breaches effectively	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Metro Rail General Rules 2020</i>	1	1	-	0.5
PC37. interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols	-	-	-	-
PC38. follow operational procedures for maintenance, station control duties, and driverless train systems	-	-	-	-
PC39. ensure compliance with rules for safe and efficient train operations	-	-	-	-
<i>Safety Circulars and Contingency Plans</i>	1	1	-	0.5
PC40. implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation	-	-	-	-
PC41. follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises	-	-	-	-
PC42. ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods	-	-	-	-
PC43. manage degraded mode operations, including train movement during interlocking failures	-	-	-	-
PC44. oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections	-	-	-	-
PC45. facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks	-	-	-	-
PC46. execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC47. oversee train stabling, depot entry/exit, train movement, and fitness checks	-	-	-	-
PC48. conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detrainment of passengers due to technical failure, explosion, or fire	-	-	-	-
PC49. operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train	-	-	-	-
PC50. coordinate vehicle movement on link lines and rescue immobilized trains	-	-	-	-
PC51. respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety	-	-	-	-
PC52. operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines	-	-	-	-
NOS Total	20	20	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N9901
NOS Name	Implement metro operations training and safety procedures
Sector	Logistics
Sub-Sector	
Occupation	Foundation – Metro (Rail)
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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DMR/LSC/N0101: Oversee Power Supply Installation (PSI) and maintenance in the traction system

Description

This unit is about managing power supply arrangements, including metering systems, energy meters, and adherence to regulatory guidelines. It covers site assessments, workforce planning, and safe operational practices for power blocks and feed extensions. Key areas include earthing fundamentals, TPC roles, interlocking principles, cable jointing, and familiarization with PSI tools and plants, aligned with Indian electricity rules and tariff regulations.

Scope

The scope covers the following :

- Power supply arrangement
- Metering systems
- Energy meters
- DERC, HERC, and UPPCL guidelines
- Site visit to RSS for supply system arrangement
- Types of power block and Permit to Work (PTW)
- Workforce planning and staff coordination
- Fundamentals, types, requirements, and benefits of earthing
- Equipment earthing and mock drill for power block availing and cancellation
- Role and duties and importance of TPC
- Feed extension safety, emergency feeding conditions, and interlocking principles
- Tools and Plants used in PSI
- Power cables, cable jointing procedures, and testing
- PSI store and project site visit for tools and cables familiarization
- Indian electricity rules and acts related to tariff

Elements and Performance Criteria

Power supply arrangement

To be competent, the user/individual on the job must be able to:

- PC1.** monitor and manage power supply sources, voltage levels, and transmission paths
- PC2.** coordinate with the Operations Control Center (OCC) to ensure seamless power transitions
- PC3.** identify and troubleshoot issues with traction power supply interruptions to minimize downtimes

Metering systems

To be competent, the user/individual on the job must be able to:

- PC4.** inspect and calibrate energy meters and metering systems at appropriate intervals for accuracy
- PC5.** detect, report, and rectify metering anomalies or failures
- PC6.** conduct audits to verify meter readings and detect energy losses

Energy meters

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To be competent, the user/individual on the job must be able to:

- PC7.** operate and interpret readings from energy meters and diagnose faults
- PC8.** maintain records of energy consumption and generate necessary reports
- PC9.** engage in preventive and corrective maintenance of energy meters

DERC, HERC, and UPPCL guidelines

To be competent, the user/individual on the job must be able to:

- PC10.** adhere to the regulatory guidelines by Delhi Electricity Regulatory Commission (DERC), Haryana Electricity Regulatory Commission (HERC), and Uttar Pradesh Power Corporation Limited (UPPCL) in day-to-day operations and energy management
- PC11.** follow updated guidelines affecting traction power supply and safety protocols
- PC12.** coordinate with local authorities to adhere to legal and compliance standards

Site visit to RSS for supply system arrangement

To be competent, the user/individual on the job must be able to:

- PC13.** inspect Receiving Substations (RSS) setups for correct power supply arrangement and safety protocols
- PC14.** identify, assess, and report potential supply or safety issues
- PC15.** coordinate with maintenance teams to ensure RSS equipment meets operational standards

Types of power block and Permit to Work (PTW)

To be competent, the user/individual on the job must be able to:

- PC16.** assess, request, and monitor power blocks based on maintenance needs and operational schedules
- PC17.** issue and oversee PTWs to authorize safe working conditions
- PC18.** ensure timely availing and cancellation of power blocks as per operational requirements

Workforce planning and staff coordination

To be competent, the user/individual on the job must be able to:

- PC19.** assess workload requirements and allocate team members based on expertise
- PC20.** coordinate tasks among staff for efficient workflow
- PC21.** maintain clear communication channels with the team for task updates

Fundamentals, types, requirements, and benefits of earthing

To be competent, the user/individual on the job must be able to:

- PC22.** apply knowledge of various earthing techniques based on equipment needs
- PC23.** conduct earthing inspections to ensure compliance with safety standards

Equipment earthing and mock drill for power block availing and cancellation

To be competent, the user/individual on the job must be able to:

- PC24.** inspect earthing setups at various sites for compliance
- PC25.** conduct mock drills for safe power block availing and cancellation

Role and duties and importance of TPC

To be competent, the user/individual on the job must be able to:

- PC26.** determine the TPC responsibilities in managing power flow and addressing emergencies
- PC27.** assist and support TPC duties when required

Feed extension safety, emergency feeding conditions, and interlocking principles

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To be competent, the user/individual on the job must be able to:

PC28. coordinate safe procedures during feed extensions and emergency scenarios

PC29. apply interlocking principles to prevent conflicting power supply situations

Tools and Plants used in PSI

To be competent, the user/individual on the job must be able to:

PC30. inspect and understand the function of different PSI tools and plants

PC31. maintain and report on tool condition and usability

Power cables, cable jointing procedures, and testing

To be competent, the user/individual on the job must be able to:

PC32. identify and differentiate power cables based on specifications and applications

PC33. follow appropriate jointing procedures and perform tests like Hi-pot and SVL checks

PC34. ensure cable integrity and safety to maintain consistent power flow without interruptions

PSI store and project site visit for tools and cables familiarization

To be competent, the user/individual on the job must be able to:

PC35. conduct regular visits to PSI stores and project sites to understand equipment inventory

PC36. familiarize with the tools and types of cables in use at the project site

Indian electricity rules and acts related to tariff

To be competent, the user/individual on the job must be able to:

PC37. apply regulations regarding electricity tariffs and safe electricity usage

PC38. ensure team compliance with Indian Electricity rules in daily operations to ensure the operations remain compliant

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the relevant traction power supply systems, including transformers, switchgear, and distribution layouts

KU2. the emergency power supply protocols and redundancy systems

KU3. the types, functions, and configurations of relevant energy meters

KU4. the metering accuracy standards, calibration techniques, and troubleshooting

KU5. the DERC, HERC, and UPPCL guidelines on power supply, tariffs, and safety standards

KU6. the legal implications and compliance requirements related to electricity usage and safety

KU7. the RSS layout, functions, and equipment, including transformers and circuit breakers

KU8. the process of RSS inspections and troubleshooting power supply issues

KU9. the electrical safety standards for human protection and equipment safety

KU10. the types of PPE required and first-aid protocols for electrical incidents

KU11. the power block processes, PTW issuance, and compliance with safety procedures

KU12. resource allocation strategies and interpersonal skills for effective coordination

KU13. the TPC duties, emergency protocols, and communication requirements with OCC

KU14. the earthing types, benefits, and safety regulations

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- KU15.** the principles of interlocking, feed extension protocols, and risk assessment for emergencies
- KU16.** Indian Electricity laws, compliance standards, and their relevance to DMRC operations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** monitor performance, track equipment conditions, and manage power sources to ensure operational efficiency
- GS2.** record data accurately and prepare reports on power consumption, equipment status, and resource allocation strategies and interpersonal skills for effective coordination compliance to support decision-making
- GS3.** liaise with internal and external stakeholders, including the OCC and regulatory authorities for smooth operations
- GS4.** coordinate with maintenance teams and contractors for efficient task delegation and clear communication
- GS5.** diagnose and troubleshoot power interruptions and equipment anomalies, applying problem-solving techniques to minimize downtime
- GS6.** ensure activities align with relevant regulations and standards, staying updated on local and regional guidelines
- GS7.** conduct inspections, audits, and calibration checks to maintain equipment accuracy, operational safety, and compliance
- GS8.** follow safety protocols, including PPE use and emergency mock drills, to mitigate safety risks
- GS9.** analyze data from energy meters to detect trends, diagnose faults, and make informed adjustments
- GS10.** interpret regulatory guidelines, applying them effectively to ensure compliance
- GS11.** allocate resources based on workload and expertise, providing guidance and maintaining communication for task updates

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Power supply arrangement</i>	2	2	-	1
PC1. monitor and manage power supply sources, voltage levels, and transmission paths	-	-	-	-
PC2. coordinate with the Operations Control Center (OCC) to ensure seamless power transitions	-	-	-	-
PC3. identify and troubleshoot issues with traction power supply interruptions to minimize downtimes	-	-	-	-
<i>Metering systems</i>	2	2	-	1
PC4. inspect and calibrate energy meters and metering systems at appropriate intervals for accuracy	-	-	-	-
PC5. detect, report, and rectify metering anomalies or failures	-	-	-	-
PC6. conduct audits to verify meter readings and detect energy losses	-	-	-	-
<i>Energy meters</i>	2	2	-	1
PC7. operate and interpret readings from energy meters and diagnose faults	-	-	-	-
PC8. maintain records of energy consumption and generate necessary reports	-	-	-	-
PC9. engage in preventive and corrective maintenance of energy meters	-	-	-	-
<i>DERC, HERC, and UPPCL guidelines</i>	2	2	-	1
PC10. adhere to the regulatory guidelines by Delhi Electricity Regulatory Commission (DERC), Haryana Electricity Regulatory Commission (HERC), and Uttar Pradesh Power Corporation Limited (UPPCL) in day-to-day operations and energy management	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow updated guidelines affecting traction power supply and safety protocols	-	-	-	-
PC12. coordinate with local authorities to adhere to legal and compliance standards	-	-	-	-
<i>Site visit to RSS for supply system arrangement</i>	1	1	-	1
PC13. inspect Receiving Substations (RSS) setups for correct power supply arrangement and safety protocols	-	-	-	-
PC14. identify, assess, and report potential supply or safety issues	-	-	-	-
PC15. coordinate with maintenance teams to ensure RSS equipment meets operational standards	-	-	-	-
<i>Types of power block and Permit to Work (PTW)</i>	2	2	-	0.5
PC16. assess, request, and monitor power blocks based on maintenance needs and operational schedules	-	-	-	-
PC17. issue and oversee PTWs to authorize safe working conditions	-	-	-	-
PC18. ensure timely availing and cancellation of power blocks as per operational requirements	-	-	-	-
<i>Workforce planning and staff coordination</i>	1	1	-	0.5
PC19. assess workload requirements and allocate team members based on expertise	-	-	-	-
PC20. coordinate tasks among staff for efficient workflow	-	-	-	-
PC21. maintain clear communication channels with the team for task updates	-	-	-	-
<i>Fundamentals, types, requirements, and benefits of earthing</i>	1	1	-	0.5
PC22. apply knowledge of various earthing techniques based on equipment needs	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. conduct earthing inspections to ensure compliance with safety standards	-	-	-	-
<i>Equipment earthing and mock drill for power block availing and cancellation</i>	1	1	-	0.5
PC24. inspect earthing setups at various sites for compliance	-	-	-	-
PC25. conduct mock drills for safe power block availing and cancellation	-	-	-	-
<i>Role and duties and importance of TPC</i>	1	1	-	0.5
PC26. determine the TPC responsibilities in managing power flow and addressing emergencies	-	-	-	-
PC27. assist and support TPC duties when required	-	-	-	-
<i>Feed extension safety, emergency feeding conditions, and interlocking principles</i>	1	1	-	0.5
PC28. coordinate safe procedures during feed extensions and emergency scenarios	-	-	-	-
PC29. apply interlocking principles to prevent conflicting power supply situations	-	-	-	-
<i>Tools and Plants used in PSI</i>	1	1	-	0.5
PC30. inspect and understand the function of different PSI tools and plants	-	-	-	-
PC31. maintain and report on tool condition and usability	-	-	-	-
<i>Power cables, cable jointing procedures, and testing</i>	1	1	-	0.5
PC32. identify and differentiate power cables based on specifications and applications	-	-	-	-
PC33. follow appropriate jointing procedures and perform tests like Hi-pot and SVL checks	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. ensure cable integrity and safety to maintain consistent power flow without interruptions	-	-	-	-
<i>PSI store and project site visit for tools and cables familiarization</i>	1	1	-	0.5
PC35. conduct regular visits to PSI stores and project sites to understand equipment inventory	-	-	-	-
PC36. familiarize with the tools and types of cables in use at the project site	-	-	-	-
<i>Indian electricity rules and acts related to tariff</i>	1	1	-	0.5
PC37. apply regulations regarding electricity tariffs and safe electricity usage	-	-	-	-
PC38. ensure team compliance with Indian Electricity rules in daily operations to ensure the operations remain compliant	-	-	-	-
NOS Total	20	20	-	10

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0101
NOS Name	Oversee Power Supply Installation (PSI) and maintenance in the traction system
Sector	Logistics
Sub-Sector	
Occupation	Traction
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DMR/LSC/N0102: Oversee power system protection and control

Description

This unit focuses on maintaining the safety, reliability, and efficient management of electrical systems in traction power operations. Key competencies include educating personnel on safety protocols, conducting risk assessments, operating equipment, troubleshooting issues, and ensuring compliance with regulatory standards through coordinated efforts with internal teams and external authorities.

Scope

The scope covers the following :

- Electrical safety
- Safe visit to the substation and working on PSI equipment
- Liaison with higher authorities and Traction Power Controller (TPC)
- Site visit to OCC for TPC working
- Traction and auxiliary power transformers
- Circuit Breakers (CBs), Isolators, Interrupters, and Control Supply of CBs
- Circuit breaker model
- Visit of circuit breaker model at the training institute
- Protection systems (O/C & E/F, Distance, REF, etc.)
- Relay types and features
- Current Transformer (CT), Potential Transformer (PT), and Lightning Arrestor (LA) applications
- Relay checks and fault data analysis at substations
- Failure analysis of PSI equipment and preventive actions
- Routine documentation and maintenance monitoring
- Gas Insulated Switchgear (GIS) system, features, and maintenance
- Meggering, earth resistance testing, BDV, and DGA tests

Elements and Performance Criteria

Electrical safety

To be competent, the user/individual on the job must be able to:

- PC1.** educate personnel on electrical safety protocols and emergency procedures
- PC2.** implement and oversee safety checks for equipment to prevent hazards
- PC3.** ensure the use of Personal Protective Equipment (PPE) and compliance with safety guidelines

Safe visit to substation and working on PSI equipment

To be competent, the user/individual on the job must be able to:

- PC4.** conduct risk assessments before starting work on PSI equipment
- PC5.** oversee adherence to lockout-tagout procedures and safe distancing
- PC6.** check that all personnel on-site follow standard operating procedures

Liaison with higher authorities and Traction Power Controller (TPC)

To be competent, the user/individual on the job must be able to:

- PC7.** report the operational status and highlight any critical issues

Qualification Pack

PC8. collaborate with TPC for smooth operational flow, e.g. during high-demand scenarios

PC9. ensure updates on power supply, emergency scenarios, and maintenance

Site visit to OCC for TPC working

To be competent, the user/individual on the job must be able to:

PC10. observe TPC's role in controlling and monitoring power distribution

PC11. familiarize with OCC workflows, coordination, and emergency handling procedures

Traction and auxiliary power transformers

To be competent, the user/individual on the job must be able to:

PC12. identify the key components and their roles in traction and auxiliary transformers

PC13. conduct inspections for maintenance and ensure operational integrity

Circuit Breakers (CBs), Isolators, Interrupters, and Control Supply of CBs

To be competent, the user/individual on the job must be able to:

PC14. identify and operate different types of circuit breakers and isolators used in traction systems

PC15. troubleshoot and oversee the control supply operations for circuit breakers

PC16. conduct inspections and ensure the effective functioning of all CB components

Visit of circuit breaker model at the training institute

To be competent, the user/individual on the job must be able to:

PC17. comprehend CB mechanisms and configurations by attending model-based training sessions

PC18. handle and maintain CBs effectively in operational settings

PC19. attend training sessions or demonstrations on the circuit breaker model at the designated training

PC20. conduct operational tests on the model and identify potential issues, noting how circuit breakers ensure safe power distribution

Protection systems (O/C & E/F, Distance, REF, etc.)

To be competent, the user/individual on the job must be able to:

PC21. identify and operate over-current (O/C), earth fault (E/F), distance protection, and restricted earth fault (REF) protections

PC22. monitor and manage relay functions for these protections to prevent faults

Relay types and features

To be competent, the user/individual on the job must be able to:

PC23. identify different types of relays and their specific functions within protection systems

PC24. adjust relay settings based on operational requirements and safety protocols

Current Transformer (CT), Potential Transformer (PT), and Lightning Arrestor (LA) applications

To be competent, the user/individual on the job must be able to:

PC25. install and configure CTs and PTs in alignment with system requirements

PC26. implement and inspect lightning arrestors (LA) to protect equipment from power surges

Relay checks and fault data analysis at substations

To be competent, the user/individual on the job must be able to:

PC27. perform relay checks and familiarize with relay setups at substations

PC28. retrieve and analyze fault data from relays and adjust settings as needed

Failure analysis of PSI equipment and preventive actions

Qualification Pack

To be competent, the user/individual on the job must be able to:

PC29. analyze and identify root causes of PSI equipment failures

PC30. implement preventive measures and monitor corrective actions taken

Routine documentation and maintenance monitoring

To be competent, the user/individual on the job must be able to:

PC31. record and monitor maintenance activities in Daily Progress Report (DPR) and UOR reports

PC32. track daily maintenance tasks and report progress to senior management

Gas Insulated Switchgear (GIS) system, features, and maintenance

To be competent, the user/individual on the job must be able to:

PC33. inspect GIS setups to ensure proper functionality and adherence to safety standards

PC34. perform scheduled maintenance and identify any signs of wear or required repairs

Meggering, earth resistance testing, BDV, and DGA tests

To be competent, the user/individual on the job must be able to:

PC35. conduct insulation resistance testing (Meggering), earth resistance tests, Breakdown Voltage (BDV) tests, and Dissolved Gas Analysis (DGA) tests

PC36. document and report test results, identifying areas requiring maintenance or repair

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the substation safety protocols, risk mitigation, and lockout-tagout procedures

KU2. the PSI equipment safety guidelines and preventive maintenance techniques to minimize risks

KU3. the components, roles, and maintenance needs for traction and auxiliary transformers

KU4. the operating principles of various CBs, isolators, and interrupters, along with their control supplies

KU5. circuit breaker components, operating mechanisms, and their role in power safety and distribution

KU6. the operation, maintenance requirements, and troubleshooting techniques for different types of circuit breakers

KU7. different types of protection mechanisms (O/C, E/F, Distance, REF) and relay configurations to safeguard traction systems from faults

KU8. relay operations, data extraction, and settings adjustments for effective fault management and troubleshooting

KU9. CT and PT connections, types, and applications, as well as the purpose and installation of lightning arrestors for system protection

KU10. how to analyze equipment failures, understand root causes, and implement preventive measures for reliable PSI operations

KU11. the standard operating reports, UORs, and DPRs for tracking and managing daily progress in maintenance and operations

KU12. GIS systems, their key features, and maintenance requirements to ensure reliable traction power distribution

Qualification Pack

KU13. the testing protocols and interpretation of results to maintain equipment integrity and prevent malfunctions

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate safety protocols, equipment status, and operational updates to ensure effective coordination and compliance with safety standards
- GS2.** analyze and troubleshoot equipment issues systematically to minimize downtime and address operational challenges
- GS3.** conduct thorough checks of equipment, documentation, and protocols, ensuring precision and adherence to safety and operational standards
- GS4.** evaluate potential hazards, implementing risk mitigation measures, and ensuring safe practices in high-voltage environments
- GS5.** work collaboratively with maintenance teams, contractors, and external authorities to maintain seamless operations and workflow
- GS6.** identify and resolve technical faults in PSI equipment and relay systems, ensuring efficient power distribution and system reliability
- GS7.** follow relevant regulations, applying compliance requirements in daily operations to meet legal and safety standards
- GS8.** interpret data from equipment and monitoring systems to detect trends, diagnose issues, and make informed adjustments
- GS9.** implement and uphold safety practices, including PPE usage, emergency preparedness, and adherence to standard operating procedures
- GS10.** maintain accurate records of inspections, maintenance, and operational status, preparing reports for senior management review

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Electrical safety</i>	1	1	-	0.5
PC1. educate personnel on electrical safety protocols and emergency procedures	-	-	-	-
PC2. implement and oversee safety checks for equipment to prevent hazards	-	-	-	-
PC3. ensure the use of Personal Protective Equipment (PPE) and compliance with safety guidelines	-	-	-	-
<i>Safe visit to substation and working on PSI equipment</i>	1	1	-	0.5
PC4. conduct risk assessments before starting work on PSI equipment	-	-	-	-
PC5. oversee adherence to lockout-tagout procedures and safe distancing	-	-	-	-
PC6. check that all personnel on-site follow standard operating procedures	-	-	-	-
<i>Liaison with higher authorities and Traction Power Controller (TPC)</i>	1	1	-	0.5
PC7. report the operational status and highlight any critical issues	-	-	-	-
PC8. collaborate with TPC for smooth operational flow, e.g. during high-demand scenarios	-	-	-	-
PC9. ensure updates on power supply, emergency scenarios, and maintenance	-	-	-	-
<i>Site visit to OCC for TPC working</i>	1	1	-	0.5
PC10. observe TPC's role in controlling and monitoring power distribution	-	-	-	-
PC11. familiarize with OCC workflows, coordination, and emergency handling procedures	-	-	-	-
<i>Traction and auxiliary power transformers</i>	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify the key components and their roles in traction and auxiliary transformers	-	-	-	-
PC13. conduct inspections for maintenance and ensure operational integrity	-	-	-	-
<i>Circuit Breakers (CBs), Isolators, Interrupters, and Control Supply of CBs</i>	2	2	-	1
PC14. identify and operate different types of circuit breakers and isolators used in traction systems	-	-	-	-
PC15. troubleshoot and oversee the control supply operations for circuit breakers	-	-	-	-
PC16. conduct inspections and ensure the effective functioning of all CB components	-	-	-	-
<i>Visit of circuit breaker model at the training institute</i>	1	1	-	0.5
PC17. comprehend CB mechanisms and configurations by attending model-based training sessions	-	-	-	-
PC18. handle and maintain CBs effectively in operational settings	-	-	-	-
PC19. attend training sessions or demonstrations on the circuit breaker model at the designated training	-	-	-	-
PC20. conduct operational tests on the model and identify potential issues, noting how circuit breakers ensure safe power distribution	-	-	-	-
<i>Protection systems (O/C & E/F, Distance, REF, etc.)</i>	2	2	-	1
PC21. identify and operate over-current (O/C), earth fault (E/F), distance protection, and restricted earth fault (REF) protections	-	-	-	-
PC22. monitor and manage relay functions for these protections to prevent faults	-	-	-	-
<i>Relay types and features</i>	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. identify different types of relays and their specific functions within protection systems	-	-	-	-
PC24. adjust relay settings based on operational requirements and safety protocols	-	-	-	-
<i>Current Transformer (CT), Potential Transformer (PT), and Lightning Arrestor (LA) applications</i>	2	2	-	1
PC25. install and configure CTs and PTs in alignment with system requirements	-	-	-	-
PC26. implement and inspect lightning arrestors (LA) to protect equipment from power surges	-	-	-	-
<i>Relay checks and fault data analysis at substations</i>	1	1	-	0.5
PC27. perform relay checks and familiarize with relay setups at substations	-	-	-	-
PC28. retrieve and analyze fault data from relays and adjust settings as needed	-	-	-	-
<i>Failure analysis of PSI equipment and preventive actions</i>	1	1	-	0.5
PC29. analyze and identify root causes of PSI equipment failures	-	-	-	-
PC30. implement preventive measures and monitor corrective actions taken	-	-	-	-
<i>Routine documentation and maintenance monitoring</i>	1	1	-	0.5
PC31. record and monitor maintenance activities in Daily Progress Report (DPR) and UOR reports	-	-	-	-
PC32. track daily maintenance tasks and report progress to senior management	-	-	-	-
<i>Gas Insulated Switchgear (GIS) system, features, and maintenance</i>	1	1	-	0.5
PC33. inspect GIS setups to ensure proper functionality and adherence to safety standards	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. perform scheduled maintenance and identify any signs of wear or required repairs	-	-	-	-
<i>Meggering, earth resistance testing, BDV, and DGA tests</i>	1	1	-	0.5
PC35. conduct insulation resistance testing (Meggering), earth resistance tests, Breakdown Voltage (BDV) tests, and Dissolved Gas Analysis (DGA) tests	-	-	-	-
PC36. document and report test results, identifying areas requiring maintenance or repair	-	-	-	-
NOS Total	20	20	-	10

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0102
NOS Name	Oversee power system protection and control
Sector	Logistics
Sub-Sector	
Occupation	Traction
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DMR/LSC/N0103: Oversee the SCADA system operations and maintenance

Description

This unit covers the process of operating and maintaining SCADA systems in power distribution, focusing on hardware, communication protocols, event management, and system integration to ensure reliable and efficient operation.

Scope

The scope covers the following :

- SCADA system overview and hardware
- Communication arrangements in SCADA
- SCADA/OCC/SHPK site visits and hardware familiarization
- Converters, transducers, and surge arresters
- SCADA alarms and event management
- SCADA lab cards
- Substation Automation System (SAS) and integration with SCADA
- Visit to RSS for SAS familiarization
- Final PSI and SCADA testing
- OCC site visit concerning Low Voltage Switchgear (LVS) Panel and UPS

Elements and Performance Criteria

SCADA system overview and hardware

To be competent, the user/individual on the job must be able to:

- PC1.** acquire an in-depth knowledge of SCADA components, including servers, interfaces, and communication devices
- PC2.** conduct site visits to SCADA, OCC, and SHPK (Sub Head Post Kiln) for hands-on hardware familiarization

Communication arrangements in SCADA

To be competent, the user/individual on the job must be able to:

- PC3.** review communication links within SCADA, including fiber optics, Ethernet, and wireless communication setups
- PC4.** troubleshoot any issues in communication channels to maintain uninterrupted SCADA functionality

SCADA/OCC/SHPK site visits and hardware familiarization

To be competent, the user/individual on the job must be able to:

- PC5.** perform detailed assessments during site visits to SCADA control centers to understand real-time operations, hardware configurations, and overall system architecture
- PC6.** visit the OCC to observe operational workflows and data interfaces between SCADA and other critical systems for traction management
- PC7.** conduct an inspection at the Sub Head Post Kiln (SHPK) to gain hands-on experience with hardware configurations and maintenance protocols in SCADA-related environments

Converters, transducers, and surge arresters

Qualification Pack

To be competent, the user/individual on the job must be able to:

PC8. identify and inspect various converters and transducers in the traction power network

PC9. ensure the proper functioning of surge arresters to protect equipment from electrical surges

SCADA alarms and event management

To be competent, the user/individual on the job must be able to:

PC10. track and manage alarms and events generated by SCADA to detect faults and anomalies in real-time

PC11. establish protocols to address critical alarms and document incident responses

SCADA lab cards

To be competent, the user/individual on the job must be able to:

PC12. • conduct site visits to SCADA labs to examine and handle different types of cards used within
• the SCADA system

PC13. troubleshoot and replace SCADA system cards as necessary

Substation Automation System (SAS) and integration with SCADA

To be competent, the user/individual on the job must be able to:

PC14. analyze the structure and operations of the SAS system, including local SCADA systems and backup control centers

PC15. coordinate SAS and SCADA functionalities for seamless substation automation

Visit to RSS for SAS familiarization

To be competent, the user/individual on the job must be able to:

PC16. conduct a site visit to the RSS to analyze SAS installations, equipment layouts, and backup systems for enhanced operational knowledge

PC17. evaluate the integration and interoperability of SCADA and SAS at the RSS to ensure seamless monitoring and control of power systems

Final PSI and SCADA testing

To be competent, the user/individual on the job must be able to:

PC18. perform system tests on SCADA and PSI setups, noting any issues and providing feedback for improvement

PC19. document testing results and submit a detailed report for final review

OCC site visit concerning Low Voltage Switchgear (LVS) Panel and UPS

To be competent, the user/individual on the job must be able to:

PC20. conduct a focused inspection at the OCC for LVS panel and UPS setups, assessing power distribution and monitoring connections to the SCADA system

PC21. review UPS system configurations at OCC to verify support for emergency power supply and resilience in SCADA operations

PC22. implement proper maintenance protocols for LVS and UPS systems to prevent power interruptions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** SCADA architecture, hardware components, and system functionalities, including key servers, panels, and interfaces
- KU2.** the communication arrangement within SCADA, including protocols, network types, and troubleshooting techniques
- KU3.** the purpose and functions of converters, transducers, and surge arresters in maintaining power stability within SCADA
- KU4.** SCADA alarms, event management protocols, and response procedures to ensure timely identification and rectification of system issues
- KU5.** various cards used in SCADA systems, their functions, and replacement and troubleshooting techniques
- KU6.** the functionalities and integration of SAS with SCADA for effective power monitoring and control
- KU7.** PSI and SCADA testing procedures, including result documentation and providing feedback for optimal setup and operation
- KU8.** LVS panel functionalities and the role of UPS systems in providing emergency power for SCADA and other essential operations at OCC
- KU9.** SCADA site layouts and operational differences across SCADA control centers, OCC, and SHPK locations, including specific hardware and configuration requirements at each site
- KU10.** the role of RSS in the SCADA network, focusing on SAS systems, backup control integration, and emergency response capabilities
- KU11.** the procedures and best practices for integrating LVS panels and UPS systems with SCADA, including emergency protocols and troubleshooting processes at the OCC
- KU12.** the methods to evaluate and document SCADA system performance during testing, including setting benchmarks for system resilience, efficiency, and fault tolerance
- KU13.** key protocols for identifying and documenting critical observations during site visits to SCADA-related locations to enhance system operability and maintenance practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively with team members, technical support staff, and management for reporting and coordination
- GS2.** listen actively to understand technical issues and respond with appropriate guidance or action plans
- GS3.** apply logical thinking and diagnostic skills to resolve technical and communication issues
- GS4.** identify system vulnerabilities and take proactive measures to prevent potential disruptions
- GS5.** maintain accuracy in monitoring alarms, events, and testing results
- GS6.** document all inspections, tests, and incident responses precisely
- GS7.** analyze system and performance data for identifying trends and patterns
- GS8.** evaluate testing outcomes to propose improvements or modifications to SCADA and PSI setups
- GS9.** produce clear, concise documentation for troubleshooting, maintenance reports, and operational procedures

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Qualification Pack

- GS10.** ensure thorough reporting of testing outcomes and system assessments for final reviews
- GS11.** follow the safety standards applicable to SCADA, substation automation systems, site visits and hardware interaction

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>SCADA system overview and hardware</i>	2	2	-	1
PC1. acquire an in-depth knowledge of SCADA components, including servers, interfaces, and communication devices	-	-	-	-
PC2. conduct site visits to SCADA, OCC, and SHPK (Sub Head Post Kiln) for hands-on hardware familiarization	-	-	-	-
<i>Communication arrangements in SCADA</i>	2	2	-	1
PC3. review communication links within SCADA, including fiber optics, Ethernet, and wireless communication setups	-	-	-	-
PC4. troubleshoot any issues in communication channels to maintain uninterrupted SCADA functionality	-	-	-	-
<i>SCADA/OCC/SHPK site visits and hardware familiarization</i>	2	2	-	1
PC5. perform detailed assessments during site visits to SCADA control centers to understand real-time operations, hardware configurations, and overall system architecture	-	-	-	-
PC6. visit the OCC to observe operational workflows and data interfaces between SCADA and other critical systems for traction management	-	-	-	-
PC7. conduct an inspection at the Sub Head Post Kiln (SHPK) to gain hands-on experience with hardware configurations and maintenance protocols in SCADA-related environments	-	-	-	-
<i>Converters, transducers, and surge arresters</i>	2	2	-	1
PC8. identify and inspect various converters and transducers in the traction power network	-	-	-	-
PC9. ensure the proper functioning of surge arresters to protect equipment from electrical surges	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>SCADA alarms and event management</i>	2	2	-	1
PC10. track and manage alarms and events generated by SCADA to detect faults and anomalies in real-time	-	-	-	-
PC11. establish protocols to address critical alarms and document incident responses	-	-	-	-
<i>SCADA lab cards</i>	2	2	-	1
PC12. • conduct site visits to SCADA labs to examine and handle different types of cards used within • the SCADA system	-	-	-	-
PC13. troubleshoot and replace SCADA system cards as necessary	-	-	-	-
<i>Substation Automation System (SAS) and integration with SCADA</i>	2	2	-	1
PC14. analyze the structure and operations of the SAS system, including local SCADA systems and backup control centers	-	-	-	-
PC15. coordinate SAS and SCADA functionalities for seamless substation automation	-	-	-	-
<i>Visit to RSS for SAS familiarization</i>	2	2	-	1
PC16. conduct a site visit to the RSS to analyze SAS installations, equipment layouts, and backup systems for enhanced operational knowledge	-	-	-	-
PC17. evaluate the integration and interoperability of SCADA and SAS at the RSS to ensure seamless monitoring and control of power systems	-	-	-	-
<i>Final PSI and SCADA testing</i>	2	2	-	1
PC18. perform system tests on SCADA and PSI setups, noting any issues and providing feedback for improvement	-	-	-	-
PC19. document testing results and submit a detailed report for final review	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>OCC site visit concerning Low Voltage Switchgear (LVS) Panel and UPS</i>	2	2	-	1
PC20. conduct a focused inspection at the OCC for LVS panel and UPS setups, assessing power distribution and monitoring connections to the SCADA system	-	-	-	-
PC21. review UPS system configurations at OCC to verify support for emergency power supply and resilience in SCADA operations	-	-	-	-
PC22. implement proper maintenance protocols for LVS and UPS systems to prevent power interruptions	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0103
NOS Name	Oversee the SCADA system operations and maintenance
Sector	Logistics
Sub-Sector	
Occupation	Traction
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DMR/LSC/N0104: Oversee the maintenance of Overhead Electrification (OHE) systems

Description

This unit covers the technical specifications, safety protocols, and maintenance practices for overhead electrification (OHE) infrastructure, including masts, portals, cantilever systems, conductors, and power block procedures, ensuring safe and efficient operation and coordination with relevant teams.

Scope

The scope covers the following :

- Mast, portal, and TTC types and technical specifications
- Cantilever system components and specifications
- Types and specifications of conductors
- Site visits for practical familiarization
- Do's and don'ts of OHE maintenance
- Safety protocols for mainline failure rectification
- Power block procedures and safety protocols
- OHE safety protocols
- Coordination with TPC for night and failure conditions
- Maintenance vehicles (CMV/OMV), ladder trolleys, and their parts
- Neutral sections and ROCS system
- OHE maintenance, foot patrolling, and routine checks
- Inspection of overlap types and components
- Auto Tensioning Device (ATD) and jumper specifications

Elements and Performance Criteria

Mast, portal, and TTC types and technical specifications

To be competent, the user/individual on the job must be able to:

- PC1.** identify and select appropriate masts and portals based on technical requirements and site specifications
- PC2.** conduct inspections to ensure proper placement and structural integrity of masts and portals

Cantilever system components and specifications

To be competent, the user/individual on the job must be able to:

- PC3.** inspect and maintain cantilever systems, ensuring all components meet technical standards
- PC4.** adjust cantilever components as necessary to optimize alignment and support

Types and specifications of conductors

To be competent, the user/individual on the job must be able to:

- PC5.** identify conductor types, e.g. contact wire, catenary wire, etc., and conduct routine inspections
- PC6.** replace and splice conductors as required, ensuring stability and conductivity

Site visits for practical familiarization

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC7.** conduct site visits to depots to study various mast and cantilever arrangements, including mast types and their ideal placements
- PC8.** perform visits to the Operations Control Center (OCC) to understand TPC working and coordination during routine and emergency situations
- PC9.** participate in Electric Traction Unit (ETU) and SHPK visits to gain hands-on familiarity with the operation and maintenance of Current Maintenance Vehicles (CMVs)
- PC10.** visit the Key Points Depot (KPD) to observe deck areas, transition bars, and end-point anchor arrangements critical to ensuring seamless OHE operation
- PC11.** conduct visits to the OHE store to study tools and maintenance equipment, including earthing bonds and records for ETU supplies and tools provided in maintenance vehicles

Do's and don'ts of OHE maintenance

To be competent, the user/individual on the job must be able to:

- PC12.** follow the essential do's and don'ts during OHE maintenance to prevent hazards and ensure safe operation, including high-traffic areas
- PC13.** undertake the specified duties, including maintenance planning, team oversight, and immediate action during fault conditions

Safety protocols for mainline failure rectification

To be competent, the user/individual on the job must be able to:

- PC14.** adhere to established safety protocols during fault rectification on the mainline during revenue hours, minimizing disruptions and ensuring personnel safety
- PC15.** coordinate with TPC and use communication protocols effectively to manage mainline failures, especially during revenue hours, ensuring rapid restoration and public safety

Power block procedures and safety protocols

To be competent, the user/individual on the job must be able to:

- PC16.** coordinate with teams for power block authorization and release
- PC17.** follow safety protocols by properly placing discharge rods before commencing maintenance

OHE safety protocols

To be competent, the user/individual on the job must be able to:

- PC18.** oversee OHE maintenance activities, ensuring all safety guidelines are followed
- PC19.** respond to faults during revenue hours with minimal workforce and maximum safety

Coordination with TPC for night and failure conditions

To be competent, the user/individual on the job must be able to:

- PC20.** maintain communication with TPC during routine and failure conditions
- PC21.** direct teams for quick restoration during mainline failures

Maintenance vehicles (CMV/OMV), ladder trolleys, and their parts

To be competent, the user/individual on the job must be able to:

- PC22.** inspect and utilize CMVs/OMVs and ladder trolleys for OHE work
- PC23.** identify and rectify common faults in CMVs to avoid delays

Neutral sections and ROCS system

To be competent, the user/individual on the job must be able to:

- PC24.** inspect and ensure correct installation of neutral sections

Qualification Pack

PC25. familiarize with ROCS components for efficient operation during transitions

OHE maintenance, foot patrolling, and routine checks

To be competent, the user/individual on the job must be able to:

PC26. perform regular foot patrolling and visual inspections of OHE infrastructure

PC27. document findings and schedule any required maintenance actions

Inspection of overlap types and components

To be competent, the user/individual on the job must be able to:

PC28. inspect different overlap types (insulated, uninsulated, and multiple span configurations) to ensure continuity and insulation where required

PC29. assess three-span and four-span overlaps for proper alignment, insulation, and structural stability

Auto Tensioning Device (ATD) and jumper specifications

To be competent, the user/individual on the job must be able to:

PC30. inspect ATD systems and adjust settings (X and Y values) for temperature variation

PC31. ensure proper installation and alignment of jumpers to support wire tension

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the mast types, TTCs, and their technical capacities for suitable placement and OHE support

KU2. stay, bracket, register, steady tubes, and their fittings for cantilever setup and maintenance

KU3.

- different conductor types, splicing methods, and replacement procedures for continuity and
- durability

KU4. the procedure of availing and releasing power blocks, safety protocols for using discharge rods, and duties during OHE maintenance

KU5. the safety procedures for managing OHE failures on the mainline, particularly during revenue hours

KU6. TPC roles, communication protocols, and coordination for resolving failures, especially during night operations

KU7. CMV/OMV parts, ladder trolleys, and common tools used in OHE maintenance

KU8. neutral sections, ROCS components, and how they impact traction power transitions and continuity

KU9. How to perform foot patrolling, document findings, and schedule maintenance to prevent issues

KU10. ATD mechanics, jumper specifications, and how to calculate and adjust for optimal OHE tension

KU11. the technical and structural specifications of sites, including cantilever setups, anchor arrangements, and OHE equipment storage

KU12. different tools and equipment available in ETU, including records of earthing bonds and other OHE maintenance tools for routine and emergency use

KU13. the critical safety measures for mainline failure rectification, particularly the do's and don'ts to follow to minimize risk during revenue hours

Qualification Pack

- KU14.** the distinctions between insulated and uninsulated overlaps and the impact of three-span and four-span configurations on OHE continuity and performance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively with team members, TPC, and other stakeholders to ensure coordinated and safe OHE operations
- GS2.** inspect OHE components, identifying faults or potential issues that could compromise infrastructure integrity and safety
- GS3.** apply logical thinking and technical knowledge to troubleshoot OHE failures and minimize downtime
- GS4.** manage time efficiently, particularly during power blocks and in high-demand situations, to complete maintenance within designated windows
- GS5.** adhere to safety protocols, including use of discharge rods, and recognize hazards associated with OHE maintenance
- GS6.** work collaboratively with the team, especially in scenarios requiring immediate action, such as mainline failures
- GS7.** document inspection findings accurately, maintenance actions, and incidents, providing detailed reports for review
- GS8.** adjust to varied working conditions, including night operations, weather changes, and unexpected OHE faults
- GS9.** evaluate situations critically to make informed decisions during OHE inspections, maintenance, and in failure conditions

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Mast, portal, and TTC types and technical specifications</i>	2	2	-	1
PC1. identify and select appropriate masts and portals based on technical requirements and site specifications	-	-	-	-
PC2. conduct inspections to ensure proper placement and structural integrity of masts and portals	-	-	-	-
<i>Cantilever system components and specifications</i>	2	2	-	1
PC3. inspect and maintain cantilever systems, ensuring all components meet technical standards	-	-	-	-
PC4. adjust cantilever components as necessary to optimize alignment and support	-	-	-	-
<i>Types and specifications of conductors</i>	2	2	-	1
PC5. identify conductor types, e.g. contact wire, catenary wire, etc., and conduct routine inspections	-	-	-	-
PC6. replace and splice conductors as required, ensuring stability and conductivity	-	-	-	-
<i>Site visits for practical familiarization</i>	2	2	-	1
PC7. conduct site visits to depots to study various mast and cantilever arrangements, including mast types and their ideal placements	-	-	-	-
PC8. perform visits to the Operations Control Center (OCC) to understand TPC working and coordination during routine and emergency situations	-	-	-	-
PC9. participate in Electric Traction Unit (ETU) and SHPK visits to gain hands-on familiarity with the operation and maintenance of Current Maintenance Vehicles (CMVs)	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. visit the Key Points Depot (KPD) to observe deck areas, transition bars, and end-point anchor arrangements critical to ensuring seamless OHE operation	-	-	-	-
PC11. conduct visits to the OHE store to study tools and maintenance equipment, including earthing bonds and records for ETU supplies and tools provided in maintenance vehicles	-	-	-	-
<i>Do's and don'ts of OHE maintenance</i>	1	1	-	1
PC12. follow the essential do's and don'ts during OHE maintenance to prevent hazards and ensure safe operation, including high-traffic areas	-	-	-	-
PC13. undertake the specified duties, including maintenance planning, team oversight, and immediate action during fault conditions	-	-	-	-
<i>Safety protocols for mainline failure rectification</i>	1	1	-	1
PC14. adhere to established safety protocols during fault rectification on the mainline during revenue hours, minimizing disruptions and ensuring personnel safety	-	-	-	-
PC15. coordinate with TPC and use communication protocols effectively to manage mainline failures, especially during revenue hours, ensuring rapid restoration and public safety	-	-	-	-
<i>Power block procedures and safety protocols</i>	1.5	1.5	-	0.5
PC16. coordinate with teams for power block authorization and release	-	-	-	-
PC17. follow safety protocols by properly placing discharge rods before commencing maintenance	-	-	-	-
<i>OHE safety protocols</i>	1	1	-	0.5
PC18. oversee OHE maintenance activities, ensuring all safety guidelines are followed	-	-	-	-
PC19. respond to faults during revenue hours with minimal workforce and maximum safety	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Coordination with TPC for night and failure conditions</i>	1	1	-	0.5
PC20. maintain communication with TPC during routine and failure conditions	-	-	-	-
PC21. direct teams for quick restoration during mainline failures	-	-	-	-
<i>Maintenance vehicles (CMV/OMV), ladder trolleys, and their parts</i>	1.5	1.5	-	0.5
PC22. inspect and utilize CMVs/OMVs and ladder trolleys for OHE work	-	-	-	-
PC23. identify and rectify common faults in CMVs to avoid delays	-	-	-	-
<i>Neutral sections and ROCS system</i>	1.5	1.5	-	0.5
PC24. inspect and ensure correct installation of neutral sections	-	-	-	-
PC25. familiarize with ROCS components for efficient operation during transitions	-	-	-	-
<i>OHE maintenance, foot patrolling, and routine checks</i>	1	1	-	0.5
PC26. perform regular foot patrolling and visual inspections of OHE infrastructure	-	-	-	-
PC27. document findings and schedule any required maintenance actions	-	-	-	-
<i>Inspection of overlap types and components</i>	1	1	-	0.5
PC28. inspect different overlap types (insulated, uninsulated, and multiple span configurations) to ensure continuity and insulation where required	-	-	-	-
PC29. assess three-span and four-span overlaps for proper alignment, insulation, and structural stability	-	-	-	-
<i>Auto Tensioning Device (ATD) and jumper specifications</i>	1.5	1.5	-	0.5

Logo
Not
Available



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC30. inspect ATD systems and adjust settings (X and Y values) for temperature variation	-	-	-	-
PC31. ensure proper installation and alignment of jumpers to support wire tension	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0104
NOS Name	Oversee the maintenance of Overhead Electrification (OHE) systems
Sector	Logistics
Sub-Sector	
Occupation	Traction
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on a knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per the assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practicals for every student at each examination/training centre based on this criterion.
6. To pass the Qualification Pack, every trainee should score a minimum of 60% of aggregate marks to successfully clear the assessment.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 60

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DMR/LSC/N9901.Implement metro operations training and safety procedures	20	20	0	10	50	15
DMR/LSC/N0101.Oversee Power Supply Installation (PSI) and maintenance in the traction system	20	20	-	10	50	20
DMR/LSC/N0102.Oversee power system protection and control	20	20	-	10	50	20
DMR/LSC/N0103.Oversee the SCADA system operations and maintenance	20	20	-	10	50	20
DMR/LSC/N0104.Oversee the maintenance of Overhead Electrification (OHE) systems	20	20	-	10	50	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	120	130	0	50	300	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
QP	Qualification Pack
MC	Model Curriculum
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
OCC	Operations Control Center
PSI	Power System Installation
RSS	Receiving Substation
PTW	Permit to Work
SVL	Surge Voltage Limiter
CBs	circuit breakers
PPE	Personal Protective Equipment
CTs	Current Transformers
LAs	lightning arrestors
DGA	Dissolved Gas Analysis
OHE	Overhead Electrification
TPC	Traction Power Controller

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.